

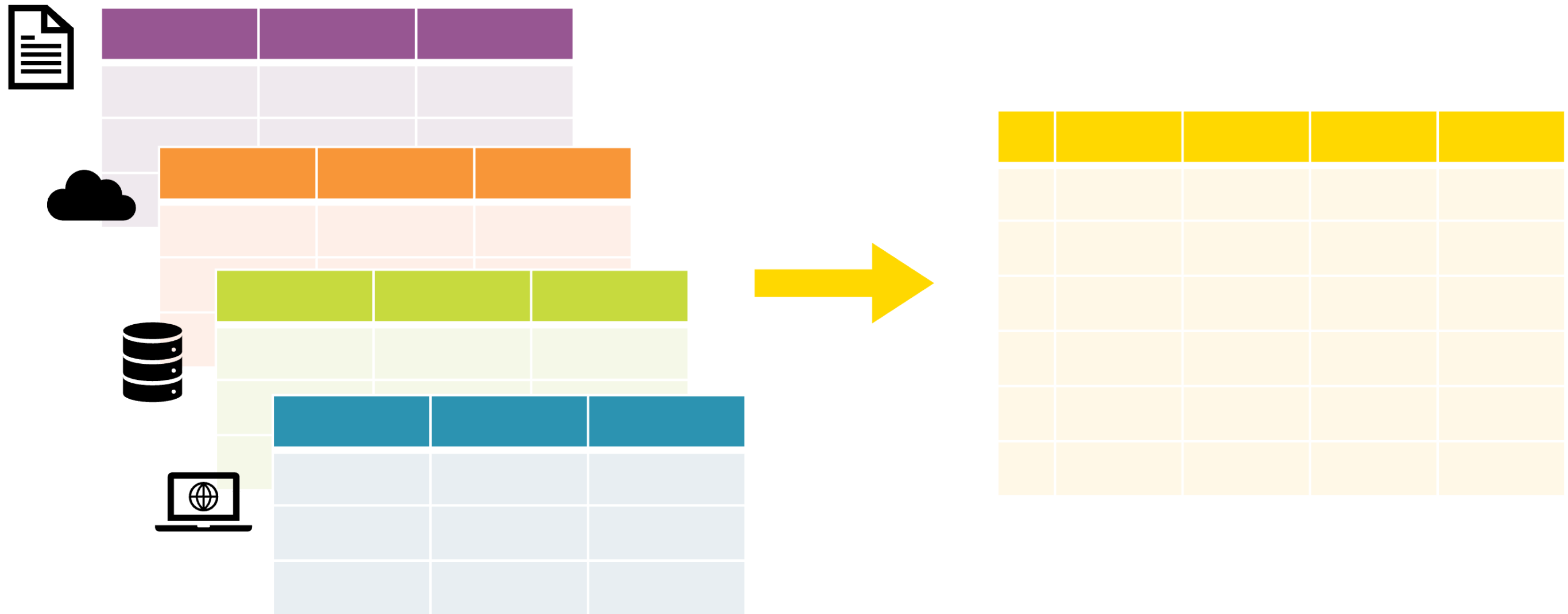
Mix and squeeze data

INTRODUCTION TO KNIME



Emilio Silvestri
Data Scientist

Data merging



Value lookup

#	EmployeeID	Name	Department	Favorite food
1	EMP-43	Jenny	Sales	Pizza
2	EMP-29	Alex	Finance	Sushi
3	EMP-11	Taylor	R&D	Tacos
4	EMP-29	Alex	Finance	Sushi
5	EMP-66	Sam	R&D	Salad

Value lookup

#	EmployeeID	Name	Department	Favorite food
1	EMP-43	Jenny	Sales	Pizza
2	EMP-29	Alex	Finance	Sushi
3	EMP-11	Taylor	R&D	Tacos
4	EMP-29	Alex	Finance	Sushi
5	EMP-66	Sam	R&D	Salad

#	ID	Salary
1	EMP-11	25000
2	EMP-29	22000
3	EMP-43	28000
4	EMP-88	22000

Value lookup

#	EmployeeID	Name	Department	Favorite food
1	EMP-43	Jenny	Sales	Pizza
2	EMP-29	Alex	Finance	Sushi
3	EMP-11	Taylor	R&D	Tacos
4	EMP-29	Alex	Finance	Sushi
5	EMP-66	Sam	R&D	Salad

#	ID	Salary
1	EMP-11	25000
2	EMP-29	22000
3	EMP-43	28000
4	EMP-88	22000

Value lookup

#	EmployeeID	Name	Department	Favorite food	Salary
1	EMP-43	Jenny	Sales	Pizza	
2	EMP-29	Alex	Finance	Sushi	
3	EMP-11	Taylor	R&D	Tacos	
4	EMP-29	Alex	Finance	Sushi	
5	EMP-66	Sam	R&D	Salad	

#	ID	Salary
1	EMP-11	25000
2	EMP-29	22000
3	EMP-43	28000
4	EMP-88	22000

Value lookup

#	EmployeeID	Name	Department	Favorite food	Salary
1	EMP-43	Jenny	Sales	Pizza	
2	EMP-29	Alex	Finance	Sushi	
3	EMP-11	Taylor	R&D	Tacos	
4	EMP-29	Alex	Finance	Sushi	
5	EMP-66	Sam	R&D	Salad	

#	ID	Salary
1	EMP-11	25000
2	EMP-29	22000
3	EMP-43	28000
4	EMP-88	22000

Value lookup

#	EmployeeID	Name	Department	Favorite food	Salary
1	EMP-43	Jenny	Sales	Pizza	28000
2	EMP-29	Alex	Finance	Sushi	
3	EMP-11	Taylor	R&D	Tacos	
4	EMP-29	Alex	Finance	Sushi	
5	EMP-66	Sam	R&D	Salad	

#	ID	Salary
1	EMP-11	25000
2	EMP-29	22000
3	EMP-43	28000
4	EMP-88	22000

Value lookup



Value lookup

#	EmployeeID	Name	Department	Favorite food	Salary
1	EMP-43	Jenny	Sales	Pizza	28000
2	EMP-29	Alex	Finance	Sushi	22000
3	EMP-11	Taylor	R&D	Tacos	25000
4	EMP-29	Alex	Finance	Sushi	22000
5	EMP-66	Sam	R&D	Salad	

#	ID	Salary
1	EMP-11	25000
2	EMP-29	22000
3	EMP-43	28000
4	EMP-88	22000

Dictionary

Value lookup

#	EmployeeID	Name	Department	Favorite food	Salary
1	EMP-43	Jenny	Sales	Pizza	28000
2	EMP-29	Alex	Finance	Sushi	22000
3	EMP-11	Taylor	R&D	Tacos	25000
4	EMP-29	Alex	Finance	Sushi	22000
5	EMP-66	Sam	R&D	Salad	

Lookup column

Key column

#	ID	Salary
1	EMP-11	25000
2	EMP-29	22000
3	EMP-43	28000
4	EMP-88	22000

Dictionary

Value lookup

#	EmployeeID	Name	Department	Favorite food	Salary
1	EMP-43	Jenny	Sales	Pizza	28000
2	EMP-29	Alex	Finance	Sushi	22000
3	EMP-11	Taylor	R&D	Tacos	25000
4	EMP-29	Alex	Finance	Sushi	22000
5	EMP-66	Sam	R&D	Salad	?

Lookup column	#	ID	Salary	Dictionary
	1	EMP-11	25000	
	2	EMP-29	22000	
	3	EMP-43	28000	
	4	EMP-88	22000	

Key
column

Data aggregation



Data aggregation



Data aggregation



#	Name	Department	Salary
1	Jenny	Sales	28000
2	Alex	Finance	22000
3	Taylor	R&D	25000
4	Andrea	Finance	22000
5	Sam	R&D	27000

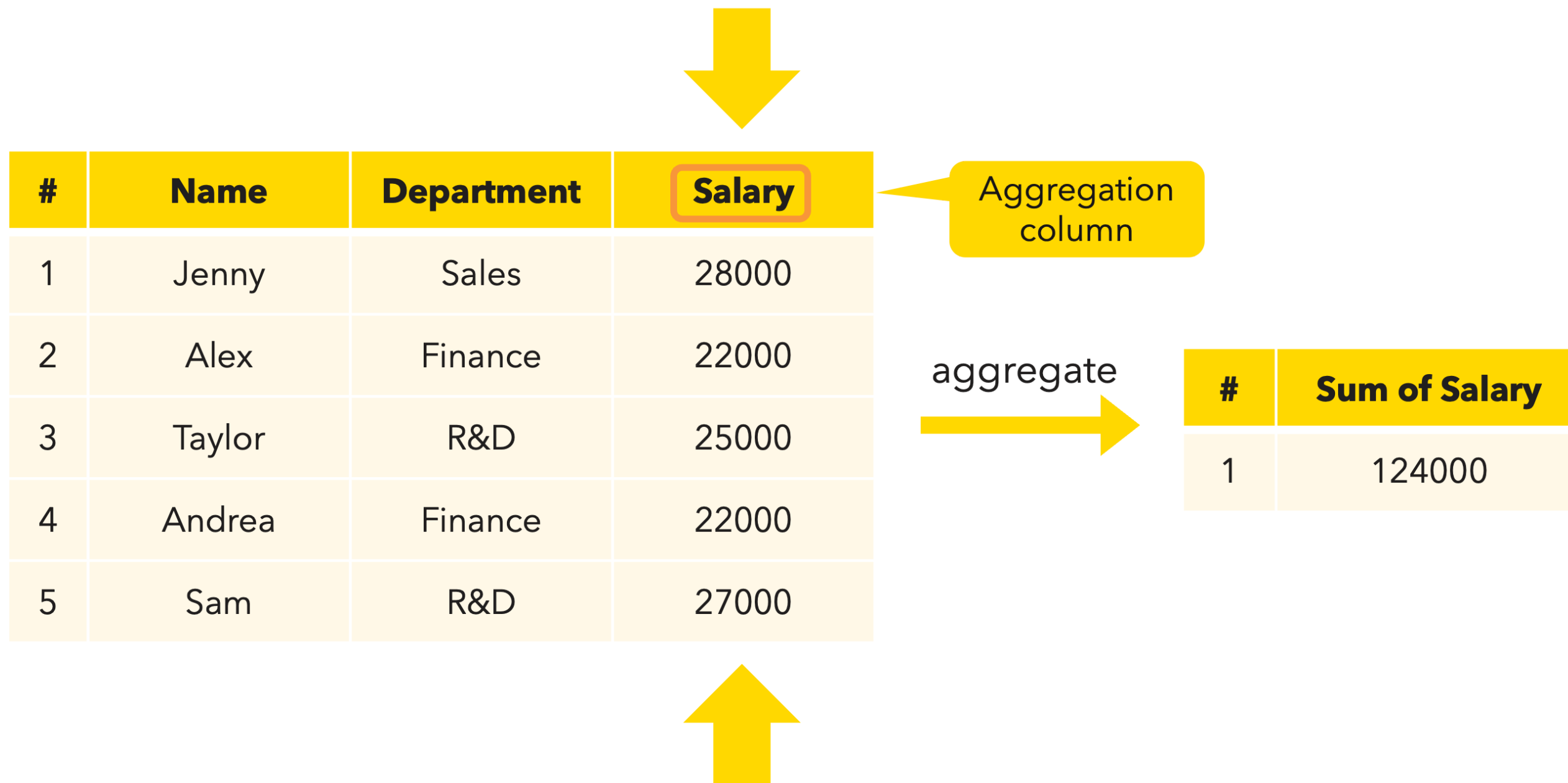
aggregate



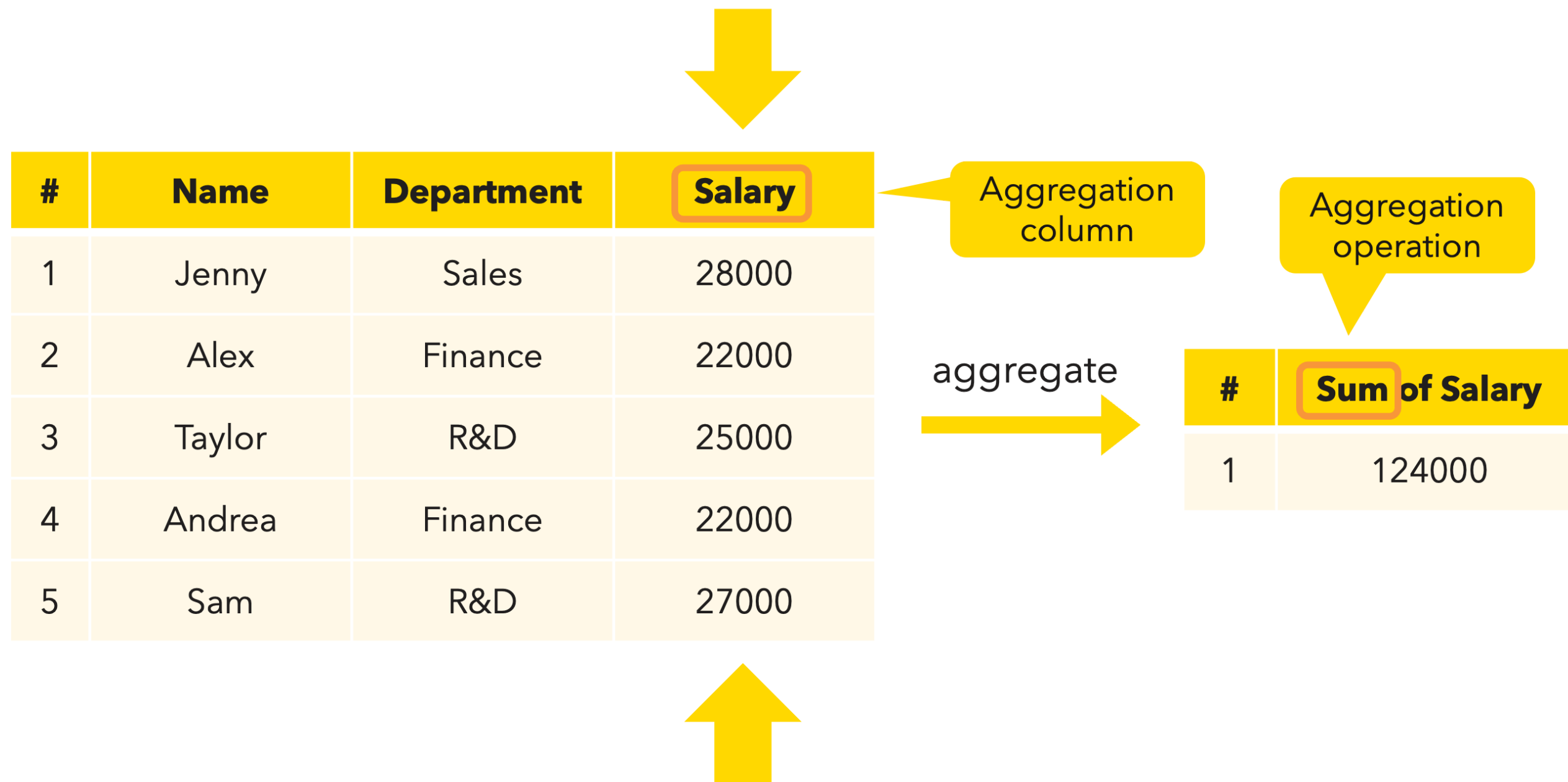
#	Sum of Salary
1	124000



Data aggregation



Data aggregation



Data aggregation

Category
column

#	Name	Department	Salary
1	Jenny	Sales	28000
2	Alex	Finance	22000
3	Taylor	R&D	25000
4	Andrea	Finance	22000
5	Sam	R&D	27000

aggregate

#	Department	Sum of Salary
1	Sales	28000
2	Finance	44000
3	R&D	52000

Data aggregation

#	Name	Department	Salary
1	Jenny	Sales	28000
2	Alex	Finance	22000
3	Taylor	R&D	25000
4	Andrea	Finance	22000
5	Sam	R&D	27000

aggregate


#	Department	Number of employees
1	Sales	1
2	Finance	2
3	R&D	2

Data aggregation

#	Name	Department	Salary
1	Jenny	Sales	28000
2	Alex	Finance	22000
3	Taylor	R&D	25000
4	Andrea	Finance	22000
5	Sam	R&D	27000

aggregate


#	Department	Average Salary
1	Sales	28000
2	Finance	22000
3	R&D	26000

Documentation



Documentation

Calculate average income

Access

Read employees dataset and income

Excel Reader



Read employees dataset

SQLite Connector



hr_data.sqlite

DB Table Selector



income table

DB Reader



Execute query

Clean

Filter and remove duplicate

Row Filter



Filter employees

Duplicate Row Filter



Remove duplicate

Column Filter



Filter columns

Merge and aggregate

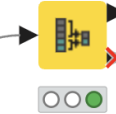
Append income and calculate average

Value Lookup



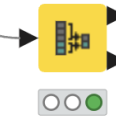
Append income

Row Aggregator



Average income in the company

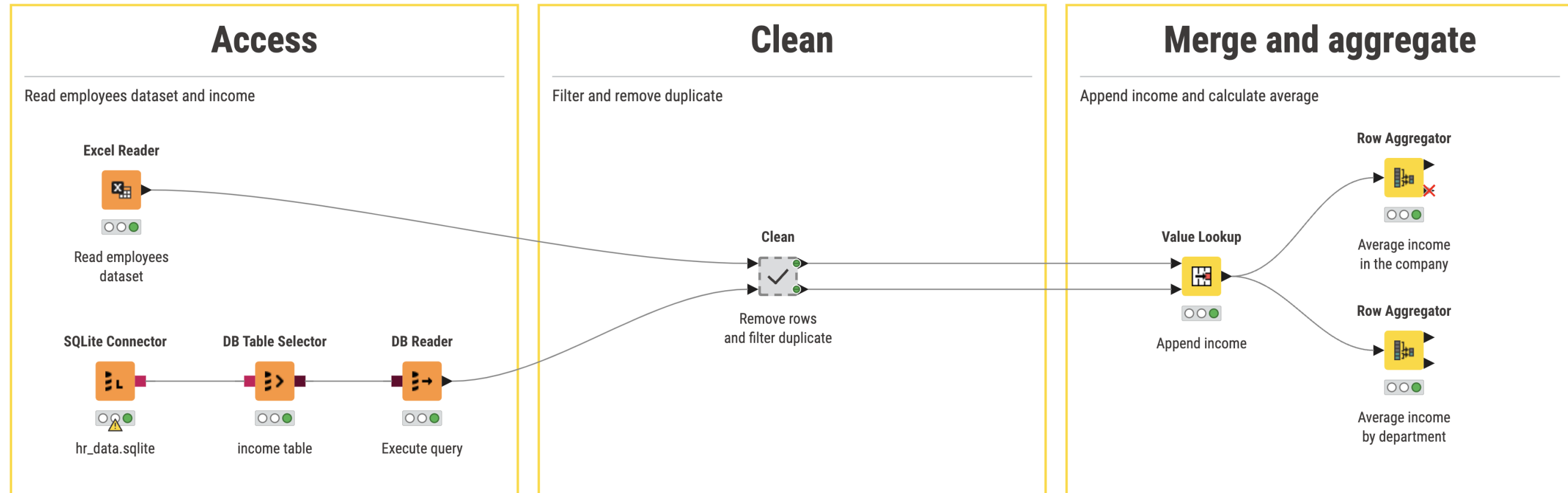
Row Aggregator



Average income by department

Metanodes

Calculate average income



Let's practice!

INTRODUCTION TO KNIME

Crunching data in KNIME Analytics Platform

INTRODUCTION TO KNIME



Emilio Silvestri
Data Scientist

Let's practice!

INTRODUCTION TO KNIME

Congratulations!

INTRODUCTION TO KNIME



Emilio Silvestri

Data Scientist, KNIME

Recap



KNIME Analytics Platform

Home

Combine Clean and Summarize Spread...

+

Help

Preferences

Menu

Execute

Cancel

Reset

Create metanode

Create component

Nodes

Search all nodes

IO

Excel Reader

Excel Writer

Microsoft Authenticator

Google Authenticator

Google Sheets Reader

Google Sheets Writer

CSV Reader

CSV Writer

Show all

Manipulation

Row Filter

Column Filter

Concatenate

Value Lookup

Row Aggregator

Table Splitter

String Cleaner

Table Cropper

Show all

Views

Bar Chart

Line Plot

Pie Chart

Stacked Area Chart

Scatter Plot

Statistics View

Heatmap

Histogram

Combine Clean and Summarize Spreadsheet Data

This workflow shows how to concatenate data from multiple Excel sheets and enrich the data with the value lookup function. At the end, it visualizes and aggregates the combined and cleaned data.

You can find more workflows and resources for spreadsheet users on the related [collection page](#) on the KNIME Community Hub.

Task

The use case is calculating the total volume of the furniture when moving to a new apartment.

Read three sheets from the same Excel file containing the following information:

- Furniture in the living room
- Furniture in the kitchen
- The estimated volumes of the furniture

Clean up the data by

- removing unnecessary columns
- shifting data that was entered into a wrong column
- changing the data type of a column

Combine data by

- concatenating rows in similar tables
- appending values from a dictionary

Calculate the following results:

- The volume by furniture type
- The total volume of all furniture

Excel Reader

Read the first sheet ("Kitchen") of the rooms.xlsx file

Excel Reader

Read the "Living Room" sheet of the rooms.xlsx file

Excel Reader

Read the "Dictionary" sheet of the rooms.xlsx file

Column Filter

Exclude the comments from the "Living Room" sheet

Column Merger

Replace missing values in the "Dict-Volume" column with the values available in the "empty_C" column

String to Number

Change "Dict-Volume" column from string to number

Concatenate

Bring the items in the "Kitchen" and "Living Room" in one table

Value Lookup

Append the volumes of the listed items based on their IDs

Bar Chart

A Bar Chart with the amounts of single items in the freight

Row Aggregator

1st output port: Sum up volumes of the listed items.
2nd output port: Calculate the grand total volume of all items.

► 1: Data Table with additional columns

Flow Variables

Table

Statistics

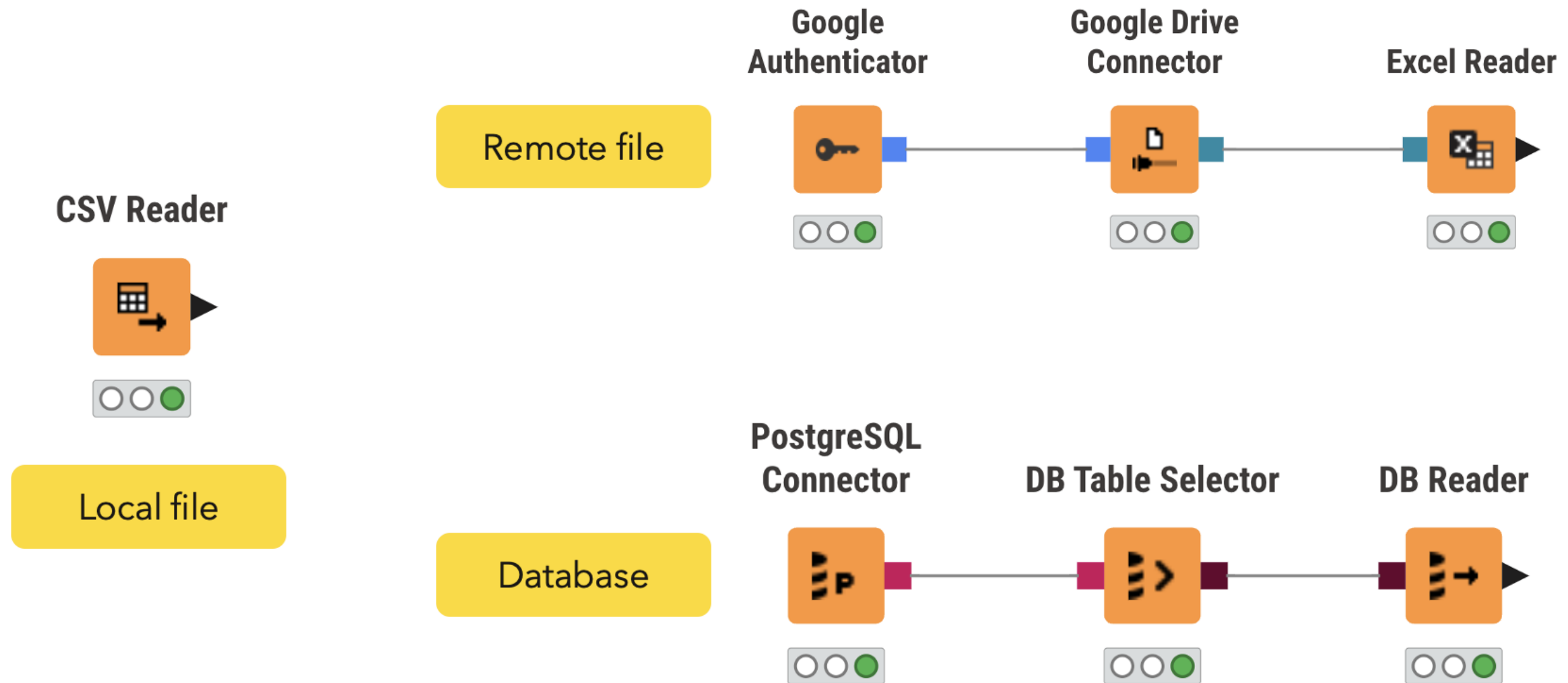
Rows: 17 | Columns: 3

#	RowID	Item	Amount	Dict-Volume
		String	Number (integer)	Number (double)
1	Row0	Table	1	0.6
2	Row1	Chair	2	0.2
3	Row2	Cupboard	1	1
4	Row3	Side table	1	1
5	Row4	Fridge	1	0.5
6	Row5	Freezer	1	1
7	Row6	Oven	1	0.2
8	Row7	Microwave	1	0.2

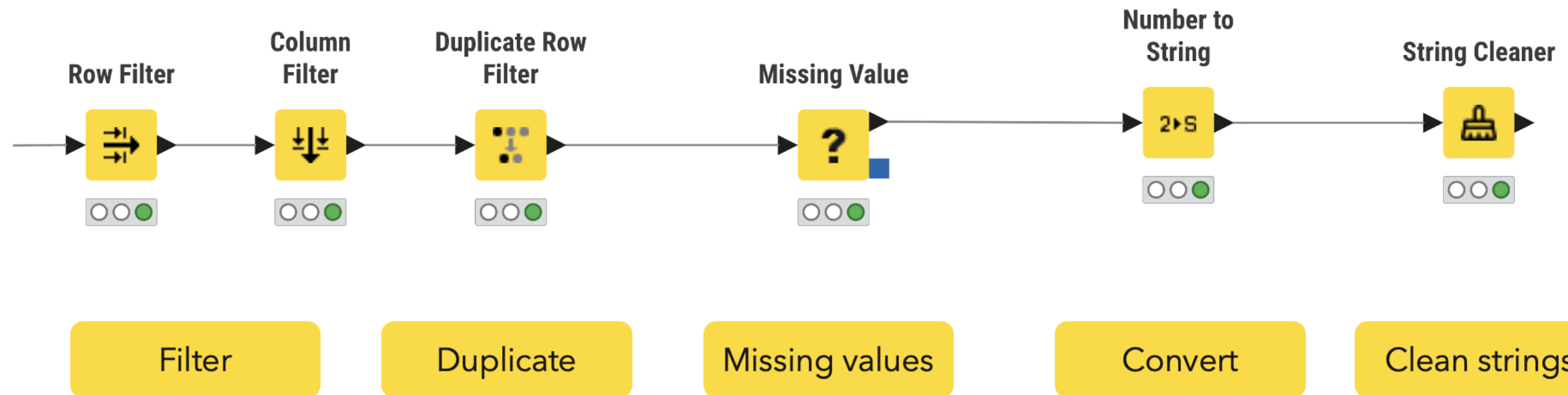
 datacamp

INTRODUCTION TO KNIME

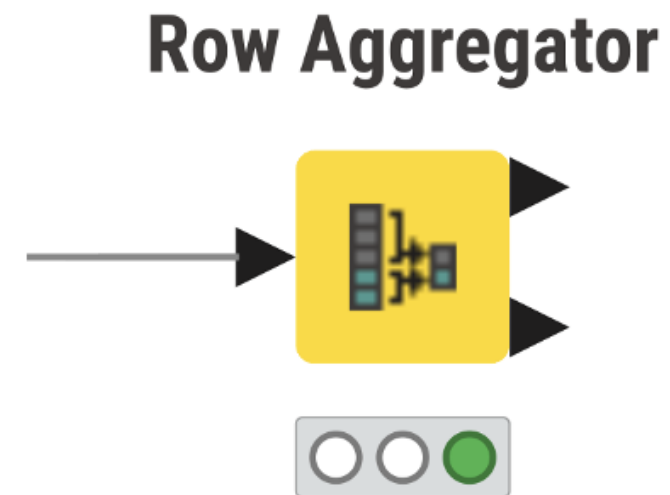
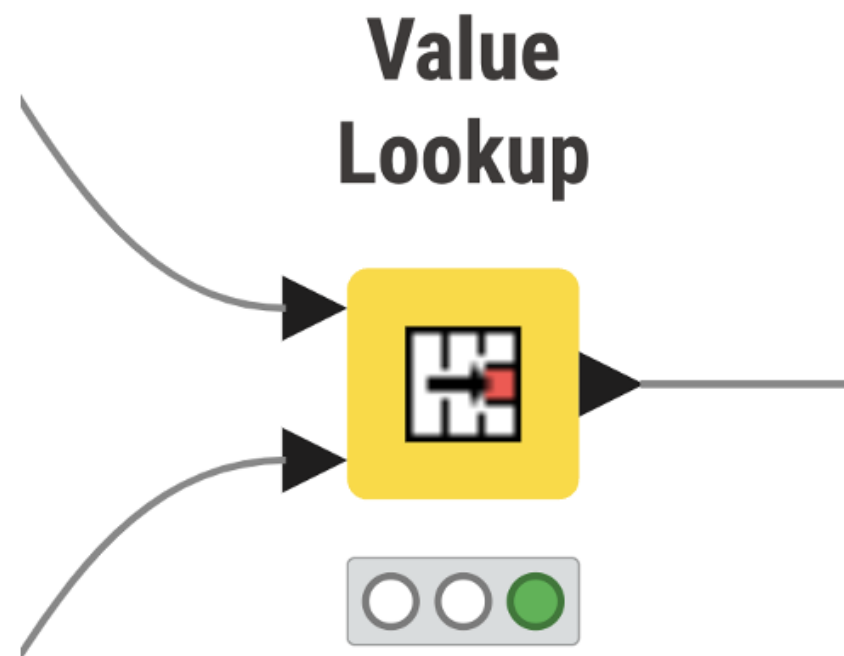
Data access



Data cleaning



Data merging and aggregation



More to come



Congratulations!

INTRODUCTION TO KNIME